

Moldova TST 2023 — P4

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Polynomials $(P_n(X))_{n \in \mathbb{N}}$ are defined as:

$$\begin{aligned} P_0(X) &= 0, & P_1(X) &= X + 2 \\ P_n(X) &= P_{n-1}(X) + 3P_{n-1}(X) \cdot P_{n-2}(X) + P_{n-2}(X), & \forall n \geq 2. \end{aligned}$$

Show that if k divides m then $P_k(X)$ divides $P_m(X)$.

Solution

One can apply Simon's favourite factoring trick to see that $3P_n + 1 = (3P_{n-1} + 1)(3P_{n-2} + 1)$. If we let $Q_n = \ln(3P_n + 1)$ then we have $Q_n = Q_{n-1} + Q_{n-2}$. Solving the recurrence relation with any standard method yields $3P_n + 1 = (3x + 7)^{f_n}$ where f_n is the n -th fibonacci number. Thus

$$P_n(x) = \frac{1}{3}[(3x + 7)^{f_n} - 1], \quad n \geq 2$$

and so the roots of P_n are of the form $\frac{1}{3}(\omega_k - 7)$, where ω_k are the f_n -th roots of unity. If $a \mid b$ then the a -th roots of unity are a proper subset of the b -th roots of unity, so we will be done if we show that $a \mid b \implies f_a \mid f_b$.

Claim 1: For non-negative integers k and m , $f_{k+m} \equiv f_m f_{k+1} \pmod{f_k}$.

Proof: We will induct on m . Throughout this proof any equivalences are assumed to be modulo f_k . If $m = 1$ then $f_{k+1} \equiv f_1 f_{k+1}$ and if $m = 2$ then $f_{k+2} \equiv f_{k+1} + f_k \equiv f_2 f_{k+1}$. This completes the base cases.

Now suppose the result holds for $m = a$ and $m = a + 1$. Consider $f_{k+a+2} \equiv f_{k+a+1} + f_{k+a} \equiv f_{k+1}(f_{a+1} + f_a) \equiv f_{k+1} f_{a+2}$. This finishes the proof by induction. $\square$

As a corollary of claim 1, we find that $f_{a+\lambda b} \equiv f_b f_{a+1}^\lambda \pmod{f_a}$ for $a, b, \lambda, \in \mathbb{Z}_{\geq 0}$. Now letting $a = b$ shows the desired result $x \mid y \implies f_x \mid f_y$. $\blacksquare$